

POLYNOMIAL GROWTH OF BOHNENBLUST–HILLE CONSTANTS ON THE HAMMING CUBE

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ABSTRACT. We prove that the Bohnenblust–Hille constants for Walsh polynomials on the Hamming cube grow at most polynomially in the degree. More precisely, there is an absolute constant K such that every $f : \{-1, 1\}^n \rightarrow \mathbb{C}$ of degree at most m satisfies

$$\left(\sum_{|S| \leq m} |\hat{f}(S)|^{2m/(m+1)} \right)^{(m+1)/(2m)} \leq K m^{27} \|f\|_{\infty}.$$

The estimate is uniform in the dimension. The exponent 27 is not optimized.

1. INTRODUCTION AND MAIN RESULTS

Let $Q_n = \{-1, 1\}^n$ be equipped with its uniform probability measure. Every function $f : Q_n \rightarrow \mathbb{C}$ has the Fourier–Walsh expansion

$$f(x) = \sum_{S \subseteq [n]} \hat{f}(S) x^S, \quad x^S = \prod_{j \in S} x_j, \quad \hat{f}(S) = \mathbb{E} f(x) x^S.$$

We say that f has degree at most m if $\hat{f}(S) = 0$ whenever $|S| > m$. Throughout the paper,

$$q_r = \frac{2r}{r+1}, \quad \frac{1}{q_r} = \frac{1}{2} + \frac{1}{2r} \quad (r \geq 1).$$

The Boolean Bohnenblust–Hille problem asks for the growth, as a function of m , of the least dimension-independent constant B_m in

$$(1.1) \quad \left(\sum_{|S| \leq m} |\hat{f}(S)|^{q_m} \right)^{1/q_m} \leq B_m \|f\|_{\infty}, \quad \deg f \leq m.$$

The classical Bohnenblust–Hille inequality arose in the solution of Bohr’s absolute convergence problem for Dirichlet series: the largest possible gap between the abscissas of absolute and uniform convergence is $1/2$ [3]. Its polynomial form relates a coefficient ℓ_{q_m} norm to a supremum norm on a polydisc. The dependence of the inequality on the degree is a central quantitative question. In the complex polynomial setting, Bayart, Pellegrino, and Seoane-Sepúlveda [1] established subexponential estimates of the form $\exp(O(\sqrt{m \log m}))$. They used these estimates to determine the precise asymptotic behavior of the Bohr radius K_n of the n -dimensional polydisc, proving $K_n \sim \sqrt{(\log n)/n}$ as $n \rightarrow \infty$ [1].

For the Boolean cube, Defant, Mastyló, and Pérez [6] proved the corresponding estimate

$$(1.2) \quad B_m \leq C \exp(c\sqrt{m \log m})$$

with absolute constants C, c .

Eskenazis and Ivanisvili [7] used these constants B_m to learn bounded low-degree functions from a number of uniform random examples logarithmic in the dimension. Related Bohnenblust–Hille inequalities also yield learning bounds for low-degree quantum observables [10].

Key words and phrases. Bohnenblust–Hille inequality, Hamming cube, Walsh polynomials, hypercontractivity, random coordinate partitions.

Recently, in a breakthrough paper, Pellegrino and Teixeira [9] obtained polynomial growth of the Bohnenblust–Hille constants for analytic polynomials on the polydisc. Their argument uses a *weighted square function*. We adapt this approach to hamming cube $\{-1, 1\}^n$. While approaches are similar there are certain distinctions between these proofs: on the polydisc, adjoining one variable makes a polynomial homogeneous without changing its coefficient or supremum norm. On the cube, this construction does not produce a homogeneous Walsh polynomial, and even a single homogeneous level $f^{\text{=r}}$ can have larger supremum norm than f . We therefore work with all degrees together in the weighted square function $S(f)$. The random coloring of [9] becomes simpler here: each variable occurs at most once in a Walsh monomial, so a binomial estimate captures, on average, a fixed proportion of every high-degree coefficient sum. Bonami–Beckner hypercontractivity bounds the resulting sums by square functions of coordinate restrictions. The weights then give a contraction at fixed dimension, while Markov’s inequality controls the finitely many low degrees.

Our main result is the following.

Theorem 1.1. *There is an absolute constant $K < \infty$ such that, for every pair of positive integers m, n with $m \leq n$ and every $f : Q_n \rightarrow \mathbb{C}$ of degree at most m ,*

$$(1.3) \quad \left(\sum_{|S| \leq m} |\hat{f}(S)|^{2m/(m+1)} \right)^{(m+1)/(2m)} \leq Km^{27} \|f\|_\infty.$$

Equivalently, $B_m \leq Km^{27}$ for every $m \geq 1$.

The proof gives the following more structured estimate. Write $f^{\text{=r}} = \sum_{|S|=r} \hat{f}(S)x^S$ and

$$\mathcal{S}_r(f) = \left(\sum_{|S|=r} |\hat{f}(S)|^{q_r} \right)^{1/q_r}.$$

Theorem 1.2. *There is an absolute constant $K < \infty$ such that every polynomial as in Theorem 1.1 satisfies*

$$(1.4) \quad \left(\sum_{r=1}^m \frac{\mathcal{S}_r(f)^2}{r^{10}} \right)^{1/2} \leq Km^{22} \|f\|_\infty.$$

In Section 2 we prove the two results together, starting with the coefficient norm in (1.3).

As an application, Theorem 1.2 establishes a special case of the Aaronson–Ambainis conjecture [12], for polynomials whose Fourier coefficients have constant absolute value on each degree level. For a real-valued function f on Q_n , write

$$\text{Var}(f) = \mathbb{E}|f - \mathbb{E}f|^2 = \sum_{S \neq \emptyset} |\hat{f}(S)|^2, \quad \text{Inf}_i(f) = \sum_{S \ni i} |\hat{f}(S)|^2.$$

Corollary 1.3. *There is an absolute constant $c > 0$ such that, for every $1 \leq m \leq n$ and every $f : Q_n \rightarrow [-1, 1]$ of degree at most m satisfying $|\hat{f}(S)| = a_{|S|}$ for some $a_0, \dots, a_m \geq 0$ and all $|S| \leq m$,*

$$\text{Inf}_i(f) \geq \frac{c}{m^{54}} \text{Var}(f)^2 \quad \text{for every } i \in [n].$$

The assumption concerns only the magnitudes of the coefficients; their signs may vary within each level. The proof of Corollary 1.3 is given in Section 3.

2. PROOF OF THE MAIN RESULTS

Proof of Theorems 1.1 and 1.2. Fix positive integers $1 \leq m \leq n$. If $f = 0$, there is nothing to prove, so by homogeneity we may assume $\|f\|_\infty = 1$. All empty sums and all layers above the degree of the polynomial are understood to be zero. In particular, after freezing coordinates, the degree bound

m may exceed the number of remaining coordinates; the same convention applies to the resulting empty layers.

We first split the coefficient sum according to degree:

$$\sum_{|S| \leq m} |\widehat{f}(S)|^{q_m} = |\widehat{f}(\emptyset)|^{q_m} + \sum_{r=1}^m \sum_{|S|=r} |\widehat{f}(S)|^{q_m}.$$

Write

$$q_r = \frac{2r}{r+1}, \quad \mathcal{S}_r(f) = \left(\sum_{|S|=r} |\widehat{f}(S)|^{q_r} \right)^{1/q_r}.$$

Since $q_r \leq q_m$, the coefficient norm on the r -th level is at most $\mathcal{S}_r(f)$. Consequently,

$$\begin{aligned} \left(\sum_{|S| \leq m} |\widehat{f}(S)|^{q_m} \right)^{1/q_m} &\leq 1 + \left(\sum_{r=1}^m \mathcal{S}_r(f)^{q_m} \right)^{1/q_m} \\ &\leq 1 + m^{1/q_m - 1/2} \left(\sum_{r=1}^m \mathcal{S}_r(f)^2 \right)^{1/2} \\ (2.1) \quad &\leq 1 + m^{5+1/(2m)} \underbrace{\left(\sum_{r=1}^m \frac{\mathcal{S}_r(f)^2}{r^{10}} \right)^{1/2}}_{S(f)}. \end{aligned}$$

The first inequality uses $|\widehat{f}(\emptyset)| \leq 1$ and the subadditivity of $t \mapsto t^{1/q_m}$, since $1/q_m \leq 1$. The second is the comparison of normalized ℓ_{q_m} and ℓ_2 norms:

$$\left(\frac{1}{m} \sum_{r=1}^m \mathcal{S}_r(f)^{q_m} \right)^{1/q_m} \leq \left(\frac{1}{m} \sum_{r=1}^m \mathcal{S}_r(f)^2 \right)^{1/2}.$$

Thus we have reduced the problem to estimating

$$S(f) = \left(\sum_{r=1}^m \frac{\mathcal{S}_r(f)^2}{r^{10}} \right)^{1/2}.$$

It remains to prove $S(f) \leq Km^{22}$. The reason for introducing the weights r^{-10} will become apparent when we split coordinates.

We may assume $m \geq 12$: for $m < 12$, the low-degree estimate below already controls all of $S(f)$. Split the weighted sum:

$$(2.2) \quad S(f) \leq \underbrace{\left(\sum_{r=1}^{11} \frac{\mathcal{S}_r(f)^2}{r^{10}} \right)^{1/2}}_{H(f): \text{ low degrees}} + \underbrace{\left(\sum_{r=12}^m \frac{\mathcal{S}_r(f)^2}{r^{10}} \right)^{1/2}}_{T(f): \text{ high degrees}}.$$

Estimating $H(f)$ is straightforward and follows from a standard fact $|f^{\neg r}(x)| \leq \frac{m^{2r}}{r!} \|f\|_\infty$. We present its proof here for completeness. For a fixed vertex x , consider

$$p_x(t) = f(tx) = \sum_{r=0}^m t^r f^{\neg r}(x).$$

Because f is multilinear, its absolute value on $[-1, 1]^n$ is bounded by its maximum on the vertices. Indeed, with all other coordinates fixed, $t \mapsto |A + Bt|$ is convex, so its maximum on $[-1, 1]$ occurs

at an endpoint. Apply this successively in each coordinate. Hence

$$|p_x(t)| \leq 1 \quad (-1 \leq t \leq 1).$$

Extracting the r -th coefficient and applying the first-derivative Markov inequality repeatedly gives (see [5, Section 5.1])

$$|f^{=r}(x)| = \frac{|p_x^{(r)}(0)|}{r!} \leq \frac{m^{2r}}{r!}.$$

For each of the fixed degrees $1 \leq r \leq 11$, use a finite homogeneous Bohnenblust–Hille constant C_r , whose existence follows from [6]:

$$\mathcal{S}_r(f) \leq C_r \|f^{=r}\|_\infty \leq \frac{C_r m^{2r}}{r!}.$$

Therefore, with a constant independent of m, n ,

$$(2.3) \quad H(f) = \left(\sum_{r=1}^{11} \frac{\mathcal{S}_r(f)^2}{r^{10}} \right)^{1/2} \leq K m^{22}, \quad \text{where} \quad K = \left(\sum_{r=1}^{11} \frac{C_r^2}{r^{10}(r!)^2} \right)^{1/2} < \infty.$$

We now need to estimate the high-degree part $T(f)$.

Keep m, n fixed and let

$$M = \sup_{\substack{\deg g \leq m \\ \|g\|_\infty \leq 1}} S(g) < \infty,$$

where g ranges over polynomials on the same n -dimensional cube; at this stage, M may depend on m, n .

We will show that, for every f with $\|f\|_\infty = 1$,

$$T(f) = \left(\sum_{r=12}^m \frac{\mathcal{S}_r(f)^2}{r^{10}} \right)^{1/2} \leq \frac{3}{4} M.$$

Together with (2.2) and (2.3), this will give

$$S(f) \leq H(f) + T(f) \leq K m^{22} + \frac{3}{4} M.$$

For $0 < \|g\|_\infty \leq 1$, apply this to $f = g/\|g\|_\infty$ and use $S(g) = \|g\|_\infty S(f)$. Thus the same bound holds for every g , such that $\|g\|_\infty \leq 1$ and $\deg(g) \leq m$. Taking the supremum over all such g 's gives

$$M \leq K m^{22} + \frac{3}{4} M,$$

and hence $M \leq 4K m^{22}$.

The class of polynomials of degree at most m is stable under freezing coordinates. Every resulting function still has degree at most m and supremum norm at most 1. A function of fewer coordinates can be regarded as a function on the original cube, independent of the unused coordinates. Its weighted coefficient norm S , the quantity on the left-hand side of (1.4), is therefore bounded by the same M .

Return to the sum

$$T(f)^2 = \sum_{r=12}^m \frac{\mathcal{S}_r(f)^2}{r^{10}}.$$

Split the coordinate indices $[n] = \{1, \dots, n\}$ randomly into two disjoint sets I and J , with $I \cup J = [n]$, assigning each index to either set with probability $1/2$, independently. Denote the resulting random partition by $\omega = (I, J)$; $\mathbb{E}_\omega$ denotes averaging over these choices. For a fixed partition, write $x = (x_i)_{i \in I}$ and $y = (y_j)_{j \in J}$; thus x now denotes only the coordinates in I , rather than the full n -dimensional vector. With this notation,

$$f(x, y) = \sum_{A \subset I, B \subset J} c_{A,B} x^A y^B, \quad c_{A,B} = \hat{f}(A \cup B).$$

Each original coefficient $\widehat{f}(S)$ occurs exactly once, with $A = S \cap I$ and $B = S \cap J$.

For a fixed set S of size r , the number of its coordinates landing in I is

$$N_S = |S \cap I| = \sum_{j \in S} \mathbf{1}_{\{j \in I\}}.$$

The indicators are independent Bernoulli variables with parameter $1/2$, so N_S has distribution $\text{Bin}(r, 1/2)$ and mean $r/2$. Hoeffding's inequality [8] therefore gives

$$\begin{aligned} p_r &:= \mathbb{P}_\omega \{r/6 \leq N_S \leq 5r/6\} \\ &= \mathbb{P}_\omega \{|N_S - r/2| \leq r/3\} \\ &= 1 - \mathbb{P}_\omega \{|N_S - r/2| > r/3\} \\ &\geq 1 - 2 \exp\left(-\frac{2(r/3)^2}{r}\right) = 1 - 2e^{-2r/9} \\ &\geq p := 1 - 2e^{-8/3} > 0.86 \quad (r \geq 12). \end{aligned}$$

The probability p_r depends only on r , and p is a common lower bound for all $r \geq 12$.

For a fixed partition $\omega = (I, J)$ and $r \geq 12$, define $C_r(\omega) \geq 0$ by

$$C_r(\omega)^{q_r} = \sum_{|S|=r} |\widehat{f}(S)|^{q_r} \mathbf{1}_{\{r/6 \leq |S \cap I| \leq 5r/6\}}.$$

Averaging the last identity gives

$$\begin{aligned} \mathbb{E}_\omega C_r(\omega)^{q_r} &= \sum_{|S|=r} |\widehat{f}(S)|^{q_r} \mathbb{P}_\omega \{r/6 \leq |S \cap I| \leq 5r/6\} \\ &= p_r \mathcal{S}_r(f)^{q_r} \geq p \mathcal{S}_r(f)^{q_r}. \end{aligned}$$

Since $q_r \leq 2$, we obtain

$$\mathcal{S}_r(f) \leq p^{-1/q_r} (\mathbb{E}_\omega C_r(\omega)^{q_r})^{1/q_r} \leq p^{-1/q_r} (\mathbb{E}_\omega C_r(\omega)^2)^{1/2}.$$

For $r \geq 12$, we also have $q_r \geq q_{12} = 24/13$, so $p^{-1/q_r} \leq p^{-13/24}$ because $0 < p < 1$. Consequently,

$$\begin{aligned} T(f) &= \left(\sum_{r=12}^m \frac{\mathcal{S}_r(f)^2}{r^{10}} \right)^{1/2} \\ &\leq \left(\sum_{r=12}^m \frac{p^{-2/q_r}}{r^{10}} \mathbb{E}_\omega C_r(\omega)^2 \right)^{1/2} \\ (2.4) \quad &\leq p^{-13/24} \left(\mathbb{E}_\omega \sum_{r=12}^m \frac{C_r(\omega)^2}{r^{10}} \right)^{1/2}. \end{aligned}$$

For a fixed partition $\omega = (I, J)$ and nonnegative integers d, e with $d + e \geq 1$, put

$$b_{d,e} = \left(\sum_{\substack{A \subset I, |A|=d \\ B \subset J, |B|=e}} |c_{A,B}|^{q_{d+e}} \right)^{1/q_{d+e}}.$$

Here d and e are the degrees in the two groups of variables; the dependence of $b_{d,e}$ on ω is implicit. On level $r \geq 12$, the *balanced pairs* are those with $d + e = r$ and $r/6 \leq d, e \leq 5r/6$. Grouping the

terms in $C_r(\omega)$ by these two degrees gives

$$C_r(\omega) = \left(\sum_{\substack{d+e=r \\ r/6 \leq d, e \leq 5r/6}} b_{d,e}^{q_r} \right)^{1/q_r}.$$

There are at most $r+1$ blocks $b_{d,e}$ in C_r , since a pair of nonnegative integers with $d+e=r$ is determined by $d \in \{0, \dots, r\}$. Since $1/q_r - 1/2 = 1/(2r)$, comparison of the ℓ_{q_r} and ℓ_2 norms of this sequence yields

$$C_r \leq (r+1)^{1/(2r)} \left(\sum_{\substack{d+e=r \\ r/6 \leq d, e \leq 5r/6}} b_{d,e}^2 \right)^{1/2}.$$

For $r \geq 12$, the prefactor satisfies $(r+1)^{1/(2r)} \leq \kappa := 13^{1/24}$. Write

$$\mathcal{B} = \{(d, e) : 12 \leq d+e \leq m, (d+e)/6 \leq d, e \leq 5(d+e)/6\}.$$

Substituting this bound for C_r into (2.4) and reindexing the sum by $r = d+e$ gives

$$\begin{aligned} T(f) &\leq p^{-13/24} \kappa \left(\mathbb{E}_\omega \sum_{r=12}^m \frac{1}{r^{10}} \sum_{\substack{d+e=r \\ r/6 \leq d, e \leq 5r/6}} b_{d,e}^2 \right)^{1/2} \\ (2.5) \quad &= p^{-13/24} \kappa \left(\mathbb{E}_\omega \sum_{(d,e) \in \mathcal{B}} \frac{b_{d,e}^2}{(d+e)^{10}} \right)^{1/2}. \end{aligned}$$

We have reached a coefficient sum over two blocks. Now comes the mixed-norm step familiar from the standard Bohnenblust–Hille argument.

For integers $d, e \geq 2$, put $r = d+e$ and write $\vartheta = d/r$. Define

$$\begin{aligned} X_{d,e} &= \left[\sum_{|A|=d} \left(\sum_{|B|=e} |c_{A,B}|^2 \right)^{q_d/2} \right]^{1/q_d}, \\ Y_{d,e} &= \left[\sum_{|B|=e} \left(\sum_{|A|=d} |c_{A,B}|^2 \right)^{q_e/2} \right]^{1/q_e}. \end{aligned}$$

Hölder and Minkowski give

$$b_{d,e} \leq X_{d,e}^{d/(d+e)} Y_{d,e}^{e/(d+e)}.$$

This is exactly Pellegrino and Teixeira's two-block coefficient inequality [9, Lemma 3.4]. We present its short proof here for completeness:

$$\begin{aligned} b_{d,e} &= \left(\sum_{|A|=d} \sum_{|B|=e} |c_{A,B}|^{q_r} \right)^{1/q_r} \\ &\stackrel{\text{Hölder}}{\leq} \left[\sum_{|A|=d} \left(\sum_{|B|=e} |c_{A,B}|^2 \right)^{q_d/2} \right]^{\vartheta/q_d} \left[\sum_{|A|=d} \left(\sum_{|B|=e} |c_{A,B}|^{q_e} \right)^{2/q_e} \right]^{(1-\vartheta)/2} \\ (2.6) \quad &\stackrel{\text{Minkowski}}{\leq} X_{d,e}^\vartheta Y_{d,e}^{1-\vartheta}. \end{aligned}$$

The first inequality applies Hölder in B and then in A ; the exponents fit because

$$\frac{1}{q_r} = \frac{\vartheta}{q_d} + \frac{1-\vartheta}{2} = \frac{\vartheta}{2} + \frac{1-\vartheta}{q_e}.$$

The second inequality uses Minkowski to bound $\|c\|_{\ell_A^2(\ell_B^{q_e})}$ by $\|c\|_{\ell_B^{q_e}(\ell_A^2)} = Y_{d,e}$, since $q_e \leq 2$.

Next, insert the *damping factors* that hypercontractivity will need later:

$$\rho_d = \sqrt{\frac{d-1}{d+1}}, \quad \hat{X}_{d,e} = \rho_d^e X_{d,e}, \quad \hat{Y}_{d,e} = \rho_e^d Y_{d,e}.$$

Every balanced pair has $d, e \geq 2$. Substituting these definitions into (2.6) gives

$$\begin{aligned} b_{d,e} &\leq \left(\frac{d+1}{d-1}\right)^{de/(2r)} \left(\frac{e+1}{e-1}\right)^{de/(2r)} \hat{X}_{d,e}^\vartheta \hat{Y}_{d,e}^{1-\vartheta} \\ (2.7) \quad &\leq 3 \hat{X}_{d,e}^\vartheta \hat{Y}_{d,e}^{1-\vartheta}. \end{aligned}$$

For the last inequality, the power-series expansion

$$k \log \frac{k+1}{k-1} = 2 \sum_{j=0}^{\infty} \frac{1}{(2j+1)k^{2j}} \quad (k > 1)$$

shows that this expression decreases with k , so for $k \geq 2$ it is at most its value $2 \log 3$ at $k = 2$. Taking the logarithm of the factor in (2.7), we obtain

$$\begin{aligned} \frac{de}{2r} \left(\log \frac{d+1}{d-1} + \log \frac{e+1}{e-1} \right) &\leq \frac{de}{2r} \left(\frac{2 \log 3}{d} + \frac{2 \log 3}{e} \right) \\ &= \log 3. \end{aligned}$$

We now match the weight $(d+e)^{-10}$ in (2.5) to the weights d^{-10} and e^{-10} , gaining a contraction factor. Divide (2.7) by r^5 :

$$\frac{b_{d,e}}{r^5} \leq 3 \frac{d^{5\vartheta} e^{5(1-\vartheta)}}{r^5} \left(\frac{\hat{X}_{d,e}}{d^5} \right)^\vartheta \left(\frac{\hat{Y}_{d,e}}{e^5} \right)^{1-\vartheta}.$$

The extra factor (recall that $\vartheta = d/r$) is

$$\frac{d^{5\vartheta} e^{5(1-\vartheta)}}{r^5} = \left[\vartheta^\vartheta (1-\vartheta)^{1-\vartheta} \right]^5.$$

For a balanced pair, $\vartheta \in [1/6, 5/6]$, and

$$\vartheta^\vartheta (1-\vartheta)^{1-\vartheta} \leq \left(\frac{5^5}{6^6} \right)^{1/6} < \frac{2}{3}.$$

Indeed, the maximum occurs at the endpoints $1/6$ and $5/6$: the logarithm $\vartheta \log \vartheta + (1-\vartheta) \log(1-\vartheta)$ is convex and symmetric.

Set

$$U_{d,e} = \frac{\hat{X}_{d,e}}{d^5}, \quad V_{d,e} = \frac{\hat{Y}_{d,e}}{e^5}.$$

Then

$$\frac{b_{d,e}}{r^5} \leq 3 \left(\frac{2}{3} \right)^5 U_{d,e}^\vartheta V_{d,e}^{1-\vartheta}.$$

Square and use weighted AM–GM, followed by $\vartheta, 1 - \vartheta \leq 1$:

$$\begin{aligned}
 \frac{b_{d,e}^2}{r^{10}} &\leq 9 \left(\frac{2}{3}\right)^{10} U_{d,e}^{2\vartheta} V_{d,e}^{2(1-\vartheta)} \\
 &\leq 9 \left(\frac{2}{3}\right)^{10} \left(\vartheta U_{d,e}^2 + (1-\vartheta) V_{d,e}^2\right) \\
 (2.8) \quad &\leq 9 \left(\frac{2}{3}\right)^{10} \left(U_{d,e}^2 + V_{d,e}^2\right).
 \end{aligned}$$

In view of (2.5) and (2.8), it remains to bound the sums of $U_{d,e}^2$ and $V_{d,e}^2$ over balanced pairs, uniformly in the partition. Since all terms are nonnegative, we may bound the larger sums over all $d, e \geq 2$.

For the first sum, the definition $U_{d,e} = d^{-5} \hat{X}_{d,e}$ gives

$$\sum_{d,e \geq 2} U_{d,e}^2 = \sum_{d \geq 2} \frac{1}{d^{10}} \sum_{e \geq 2} \hat{X}_{d,e}^2.$$

We first fix $d \geq 2$ and estimate the inner sum over e ; afterward we restore the weight d^{-10} and sum over d . Recall that

$$\hat{X}_{d,e} = \rho_d^e X_{d,e} = \left[\sum_{|A|=d} \left(\sum_{|B|=e} \rho_d^{2e} |c_{A,B}|^2 \right)^{q_d/2} \right]^{1/q_d},$$

where $\rho_d = \sqrt{(d-1)/(d+1)}$. For each $A \subset I$ with $|A| = d$, collect the entire row

$$R_A(y) = \sum_{B \subset J} c_{A,B} y^B.$$

We keep all y -degrees together here. The noise operator is $T_\rho(\sum_B u_B y^B) = \sum_B \rho^{|B|} u_B y^B$. Minkowski, Parseval, and the Bonami–Beckner hypercontractive inequality [4, 2] give the following chain:

$$\begin{aligned}
 \left(\sum_{e \geq 2} \hat{X}_{d,e}^2 \right)^{1/2} &\leq \left[\sum_{|A|=d} \left(\sum_{B \subset J} \rho_d^{2|B|} |c_{A,B}|^2 \right)^{q_d/2} \right]^{1/q_d} \\
 &= \left(\sum_{|A|=d} \|T_{\rho_d} R_A\|_2^{q_d} \right)^{1/q_d} \\
 &\leq \left(\sum_{|A|=d} \|R_A\|_{q_d}^{q_d} \right)^{1/q_d} \\
 (2.9) \quad &= \left(\mathbb{E}_y \sum_{|A|=d} |R_A(y)|^{q_d} \right)^{1/q_d}.
 \end{aligned}$$

The first step uses the direction $\ell_e^2(\ell_A^{q_d}) \leq \ell_A^{q_d}(\ell_e^2)$, valid since $q_d \leq 2$, and then includes the nonnegative terms of y -degrees 0 and 1. The hypercontractive step uses exactly $\rho_d^2 = q_d - 1$. It applies to the full, possibly inhomogeneous polynomial R_A .

For fixed y , the numbers $R_A(y)$ are the degree- d coefficients of $x \mapsto f(x, y)$. Hence the last expression in (2.9) equals

$$(\mathbb{E}_y \mathcal{S}_d(f(\cdot, y))^{q_d})^{1/q_d} \leq \left(\mathbb{E}_y \mathcal{S}_d(f(\cdot, y))^2 \right)^{1/2}.$$

Square, divide by d^{10} , and sum in d :

$$(2.10) \quad \sum_{d,e \geq 2} U_{d,e}^2 \leq \mathbb{E}_y \sum_{d \geq 2} \frac{\mathcal{S}_d(f(\cdot, y))^2}{d^{10}} \leq M^2.$$

The last step holds because $f(\cdot, y)$ has degree at most m and supremum norm at most 1. Similarly, interchanging x and y gives

$$(2.11) \quad \sum_{d,e \geq 2} V_{d,e}^2 \leq M^2.$$

This explains why we introduced M : the coefficient sums produced by hypercontractivity are precisely the weighted coefficient sums of coordinate restrictions.

Combining (2.8), (2.10), and (2.11), for every partition,

$$\sum_{(d,e) \in \mathcal{B}} \frac{b_{d,e}^2}{(d+e)^{10}} \leq 18 \left(\frac{2}{3}\right)^{10} M^2.$$

Substitute this into (2.5):

$$(2.12) \quad T(f) \leq \underbrace{p^{-13/24} \kappa 3\sqrt{2} \left(\frac{2}{3}\right)^5}_c M \leq \frac{3}{4} M$$

Remark 2.1. The balance condition $r/6 \leq d, e \leq 5r/6$ is what makes $c < 1$: it bounds the weight factor $[\vartheta^\vartheta(1-\vartheta)^{1-\vartheta}]^5$ by $(2/3)^5$, enough to absorb the other losses.

Returning to our original split (2.2) we obtain

$$M \leq Km^{22} + \frac{3}{4}M,$$

and hence $M \leq 4Km^{22}$. By scaling (restoring homogeneity in f), we obtain

$$(2.13) \quad S(f) \leq 4Km^{22} \|f\|_\infty.$$

Everything was proved at fixed n , and the resulting constant is independent of n . There is no need to assume in advance that the supremum over all dimensions is finite. For $m < 12$, the same low-degree estimate, with the sum stopped at $r = m$, already gives the required bound.

Finally, return to the coefficient norm we started with. By (2.1),

$$\begin{aligned} \left(\sum_{|S| \leq m} |\hat{f}(S)|^{q_m} \right)^{1/q_m} &\leq \left(1 + 4Km^{27+1/(2m)} \right) \|f\|_\infty \\ &\leq K'm^{27} \|f\|_\infty, \end{aligned}$$

because $m^{1/(2m)} \leq 2$. Thus the argument gives

$$\left(\sum_{|S| \leq m} |\hat{f}(S)|^{2m/(m+1)} \right)^{(m+1)/(2m)} \leq K'm^{27} \|f\|_\infty.$$

This proves both theorems. $\square$

Remark 2.2. We separate the first eleven levels so that the random-partition argument applies uniformly to the remaining degrees. With our balance interval, $r \geq 12$ ensures $d, e \geq r/6 \geq 2$; hence $q_d, q_e > 1$ and both damping factors are positive, so they can be removed at a uniformly bounded cost. The same cutoff gives the common probability bound $p = 1 - 2e^{-8/3}$ used in the contraction estimate.

Estimating the first eleven levels by Markov's inequality costs m^{22} , and removing the weight r^{-5} costs m^5 . These two contributions give the exponent 27. For simplicity of the argument, we did not try to optimize the exponent.

3. APPLICATION TO THE AARONSON–AMBAINIS CONJECTURE

The Aaronson–Ambainis conjecture [12, Conjecture 1.7] asserts that every $f : Q_n \rightarrow [-1, 1]$ of degree at most m has a coordinate i with $\text{Inf}_i(f) \geq c_0(\text{Var}(f)/m)^{C_0}$ for absolute constants $c_0, C_0 > 0$. For coefficients of one common magnitude, the implication from polynomial Bohnenblust–Hille bounds was already noted in [6]; here the magnitude may vary with the degree.

Proof of Corollary 1.3. Put $v_r = \sum_{|S|=r} |\hat{f}(S)|^2 = \binom{n}{r} a_r^2$. The hypothesis gives, for every $i \in [n]$,

$$\text{Var}(f) = \sum_{r=1}^m v_r, \quad \text{Inf}_i(f) = \sum_{r=1}^m \binom{n-1}{r-1} a_r^2 = \frac{1}{n} \sum_{r=1}^m r v_r.$$

Moreover, since $2/q_r = 1 + 1/r$ and $\binom{n}{r} \geq (n/r)^r$,

$$\mathcal{S}_r(f)^2 = \binom{n}{r}^{1/r} v_r \geq \frac{n}{r} v_r.$$

Cauchy–Schwarz and Theorem 1.2 therefore yield

$$\begin{aligned} \text{Var}(f)^2 &\leq \left(\sum_{r=1}^m r v_r \right) \left(\sum_{r=1}^m \frac{v_r}{r} \right) \\ &\leq \text{Inf}_i(f) \sum_{r=1}^m \mathcal{S}_r(f)^2 \\ &\leq m^{10} \text{Inf}_i(f) \sum_{r=1}^m \frac{\mathcal{S}_r(f)^2}{r^{10}} \\ &\leq K^2 m^{54} \text{Inf}_i(f), \end{aligned}$$

where the last step uses $\|f\|_\infty \leq 1$. Taking $c = K^{-2}$ proves the corollary. $\square$

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